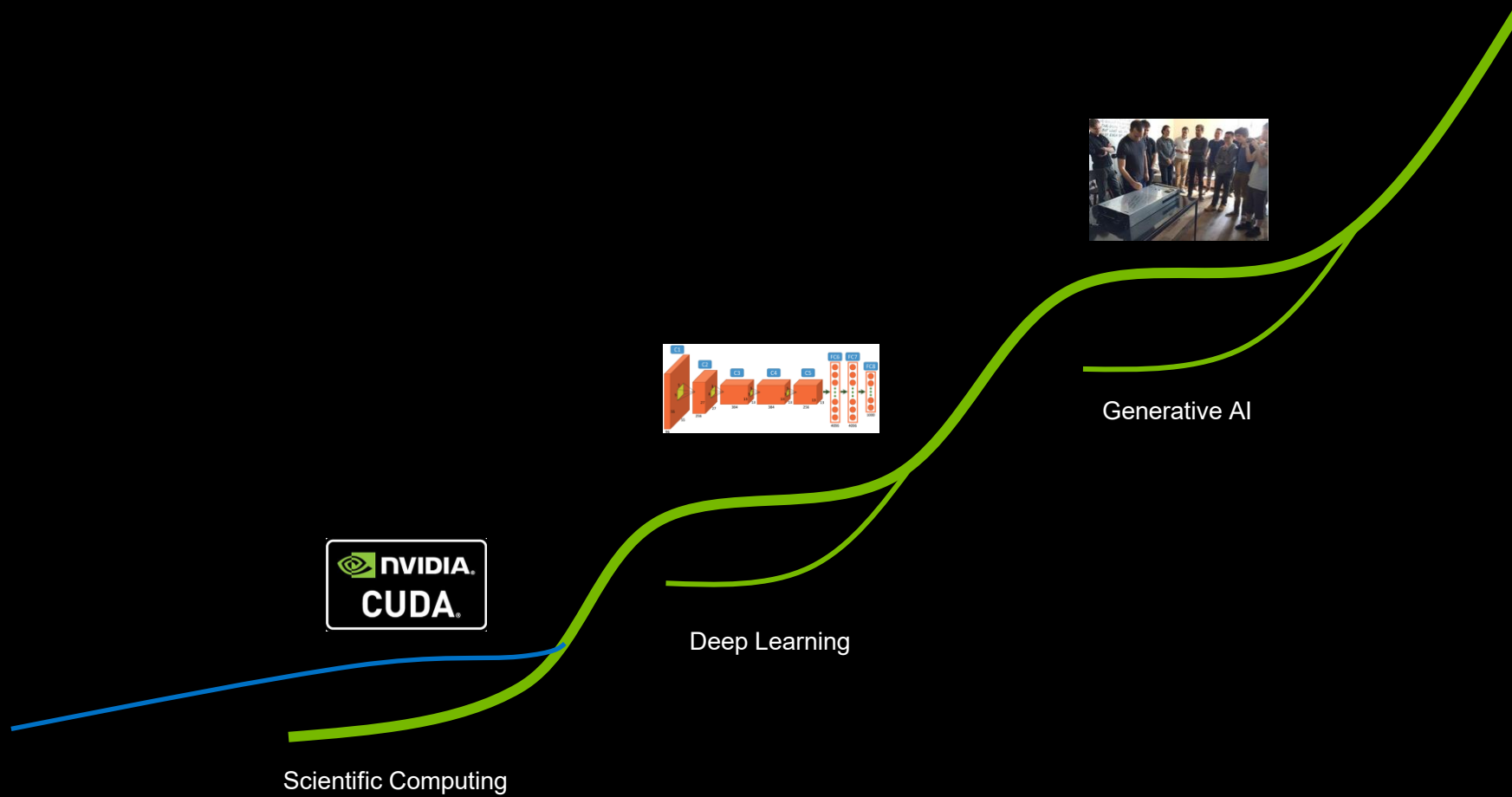


Scalable Validation and Optimized Simulation for Quantum Machine Learning

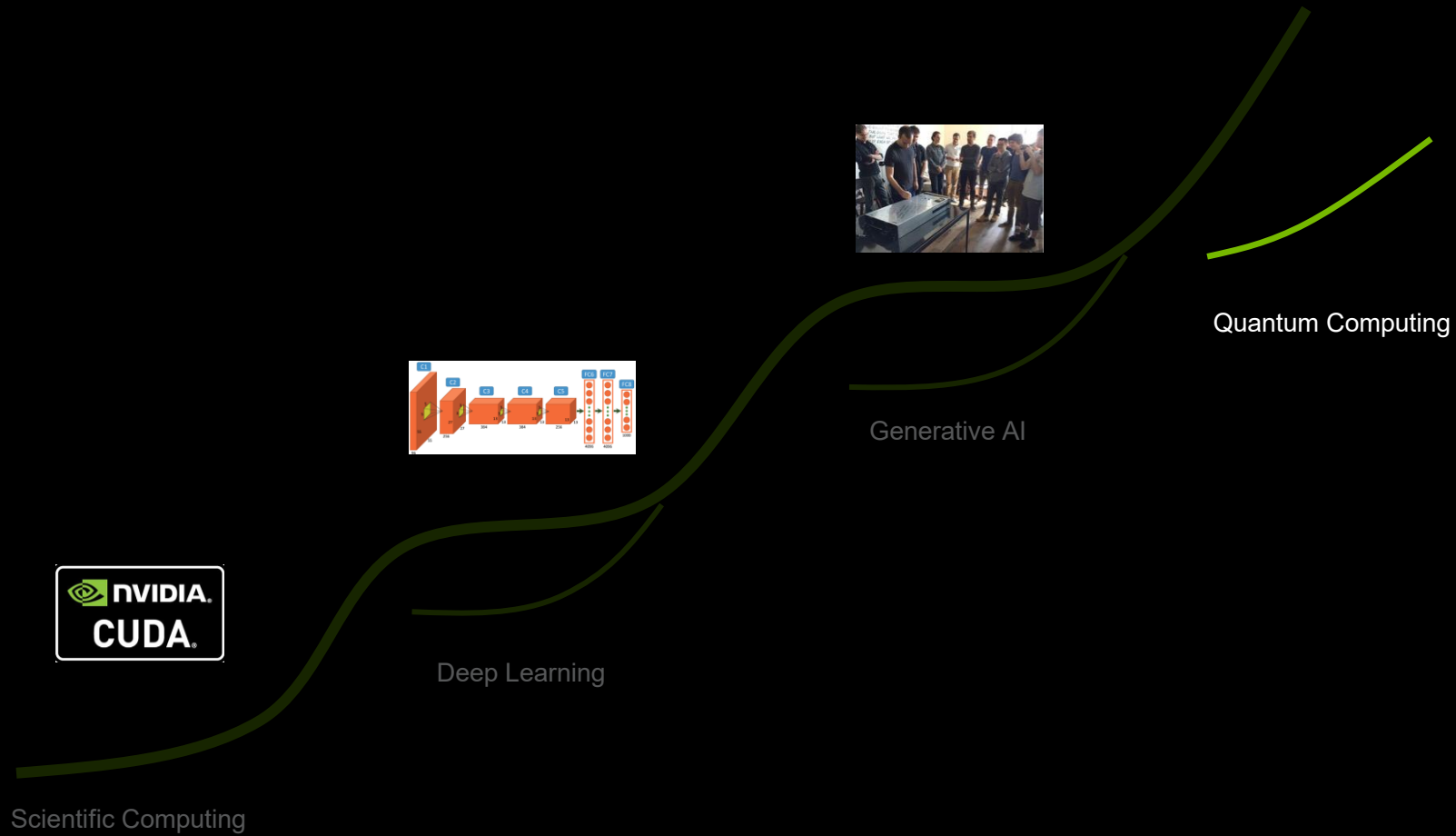
Tutorials · QCE 26

Yun Yuan (Pika) Wang, Sr. Quantum Solutions Architect | NVIDIA
Kuan-Cheng (Louis) Chen, Lead Quantum Researcher | Imperial College London / Jij Inc.
Tai-Yue Li, Sr. Researcher | NCHC
Samuel Yen-Chi Chen, Lead Research Scientist, Wells Fargo
Jiun-Cheng Jiang, Solution Architect | NVIDIA
Kevin Ho, Researcher | NCHC

NVIDIA's History of Enabling Computing Revolutions



NVIDIA's History of Enabling Computing Revolutions





NVIDIA is not building
Qubits



NVIDIA is building all
Accelerated Quantum
Supercomputers

AI and Quantum

AI to Advance Quantum Computing



Algorithms

QEC

Calibration

Digital Twins

Quantum Computing to Advance AI



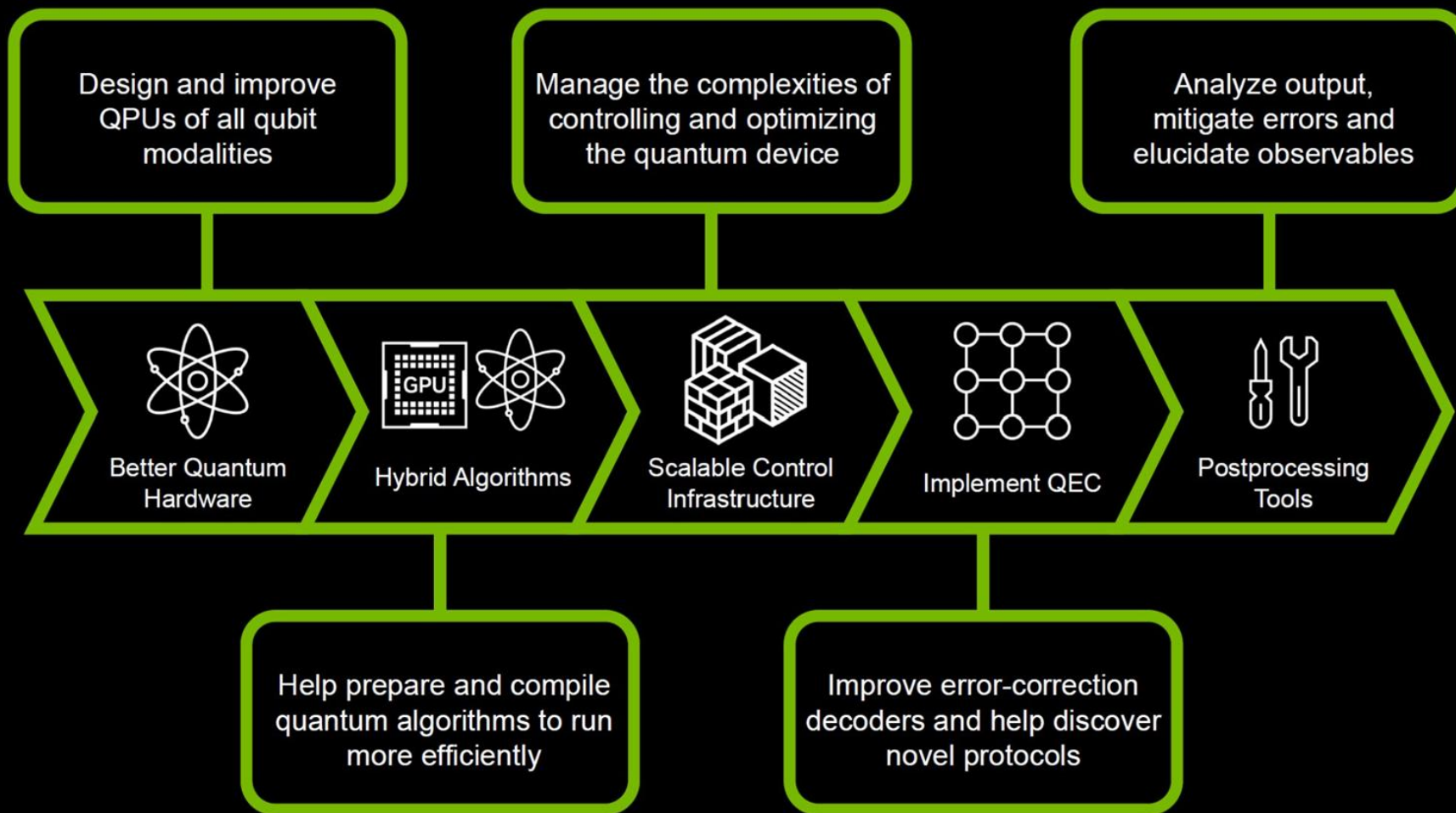
Hybrid Applications

Quantum Data
Generation

Expand Frontier of
Accelerated Computing

Artificial Intelligence for Quantum Computing

[arXiv:2411.09131](https://arxiv.org/abs/2411.09131)



The Generative Quantum Eigensolver

First demonstration of GPT-generated circuits



Challenge

- Variational quantum algorithms offered promise for running drug-discovery applications on small quantum devices - but **suffer from serious optimization issues**.
- Many of these problems are tied to how circuits are parametrized.

Solution

- The generative quantum eigensolver acts like a Large Language Model – but generating quantum circuits from quantum operations, rather than sentences from words.
- Using a generative model like GPT to create quantum circuits **avoids the limitations of traditional variational quantum algorithms**

$\vec{j} \sim$ "Once upon a time . . . happily ever after"
word_{j₁} word_{j₂} word_{j₃} word_{j₄} . . . word_{j_{N-2}} word_{j_{N-1}} word_{j_N} **LLM**

$\vec{j} \sim$
A diagram of a quantum circuit. It consists of multiple horizontal lines representing qubits. The circuit is divided into sections by vertical lines, each labeled with a gate U_j. The gates are represented by colored squares (black, green, blue, yellow) on the qubit lines. The sequence of gates is U_j1, U_j2, U_j3, followed by an ellipsis, and then U_jN. **GPT-QE**



New approach in
Using AI for building
quantum applications



Can be
extended to various
application
areas

40X

Speedup over
CPU when
running GQE on
GPU



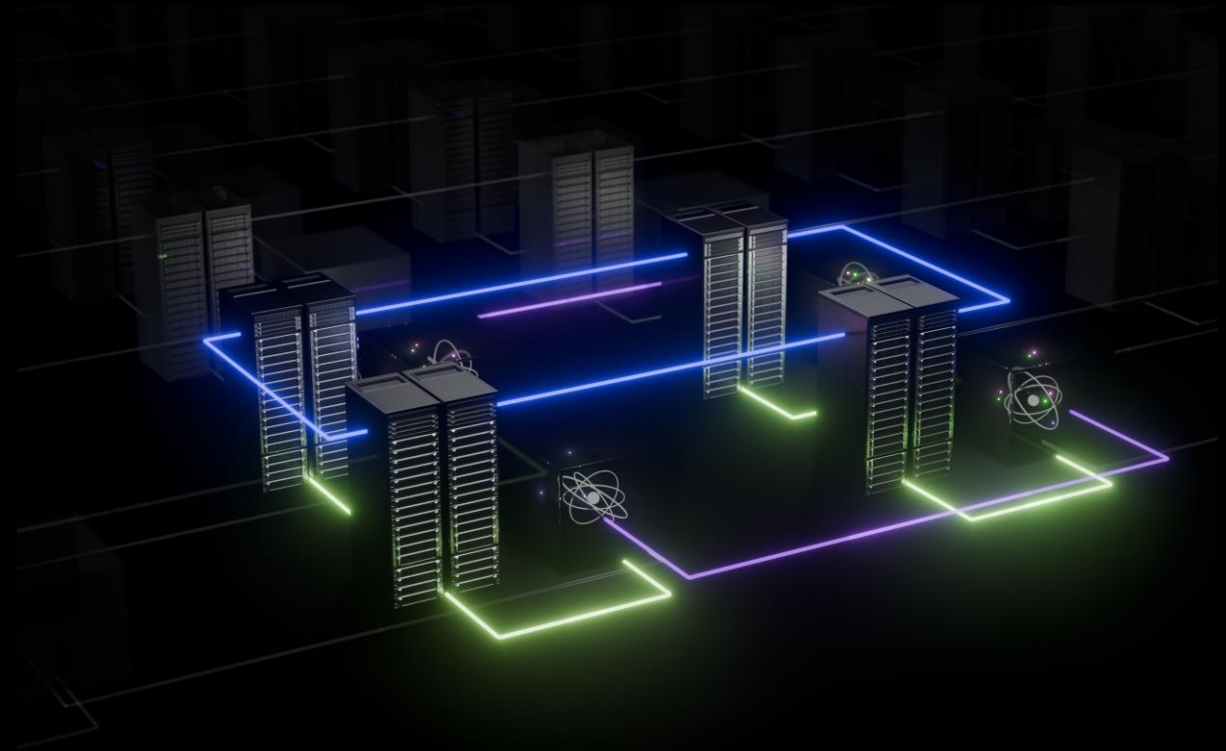
Quantum-GPU Supercomputing

All valuable quantum computing will be **hybrid**

GPUs and AI will prove essential

Petascale compute needed for useful Fault Tolerant Quantum Computers

Developer ecosystem is critical



The NVIDIA Quantum Platform

Bringing AI Supercomputing to Enable Useful Quantum Computing

CUDA-QEC and CUDA-Q Solvers

Turnkey Quantum Research and Application Development

CUDA-Q

Open-Source Software Platform for Accelerated Quantum Supercomputers

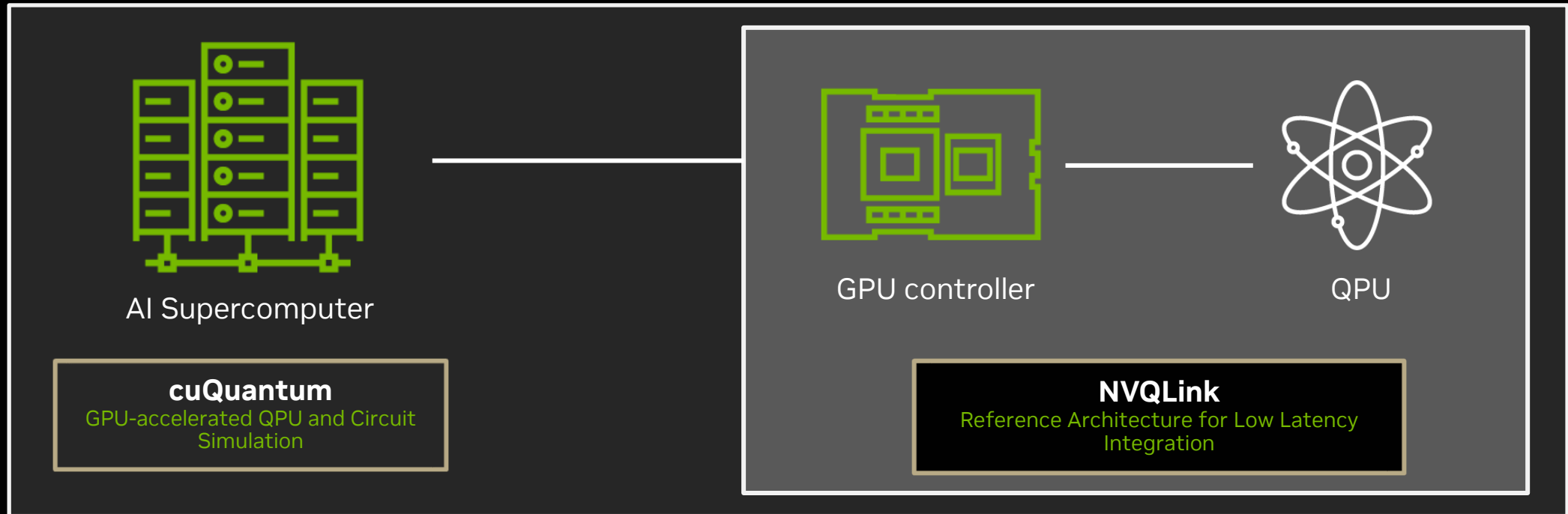
cuQuantum

Accelerated Quantum Simulation/Emulation on GPU

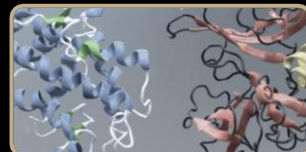
NVQLink

Reference Architecture for Low Latency Integration

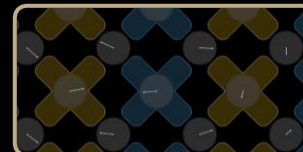
NVIDIA Open Quantum Platform



CUDA-Q
Quantum-GPU Platform

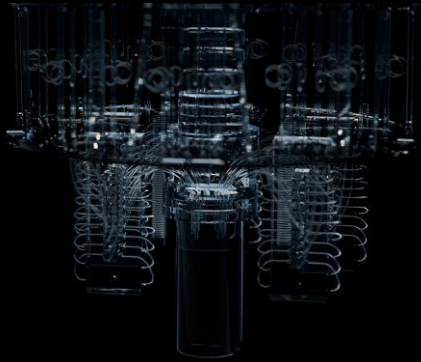


CUDA-Q Solvers

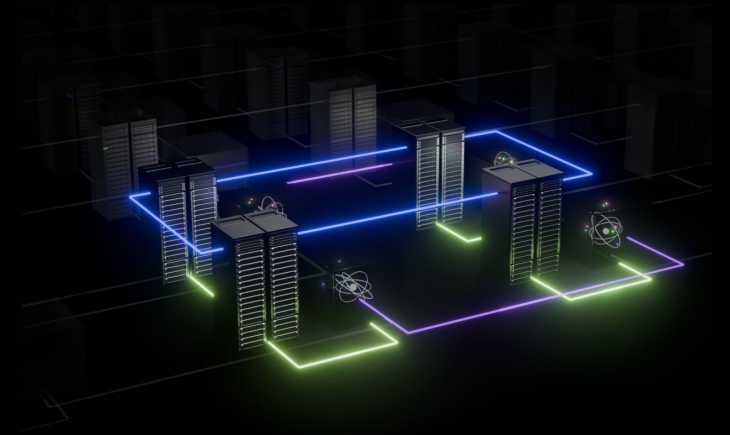


CUDA-Q QEC

The Path from Qubits to Supercomputers

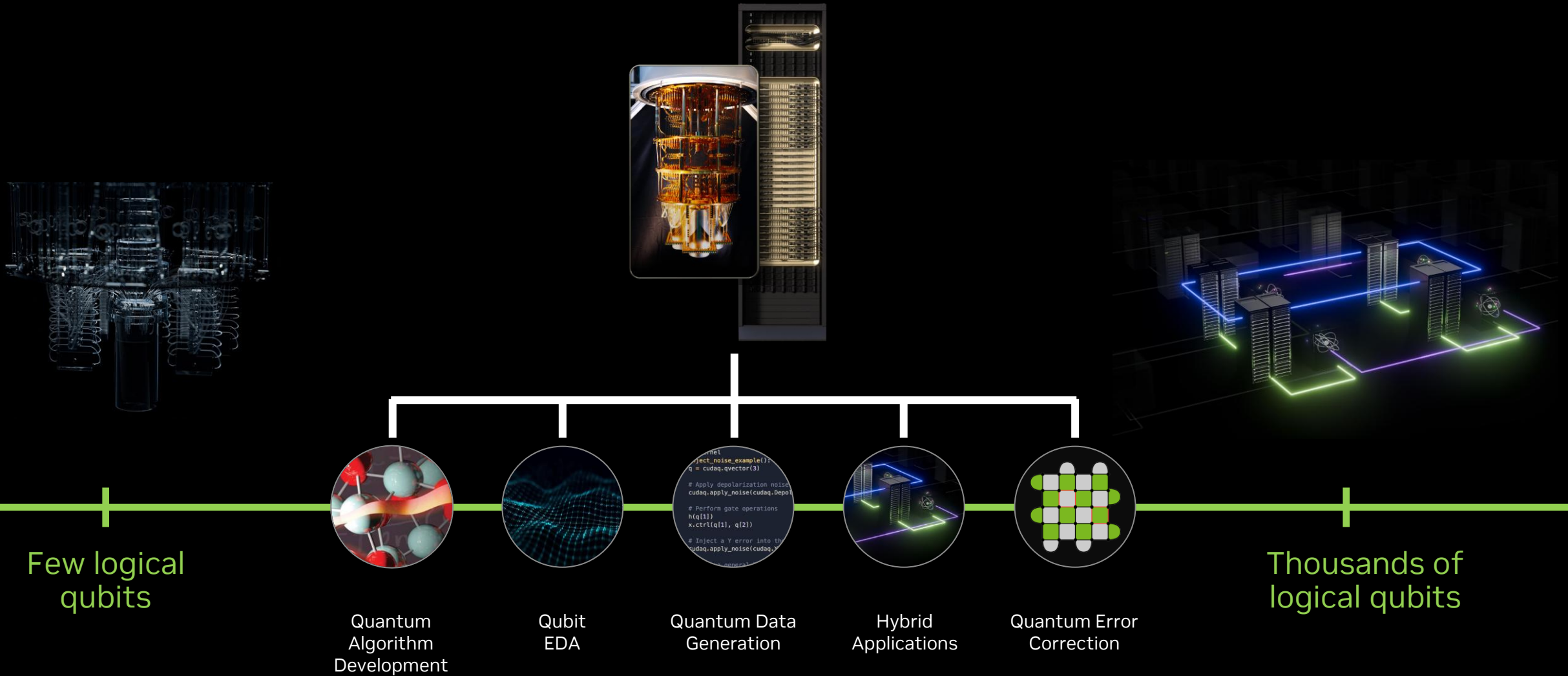


Today's
Systems

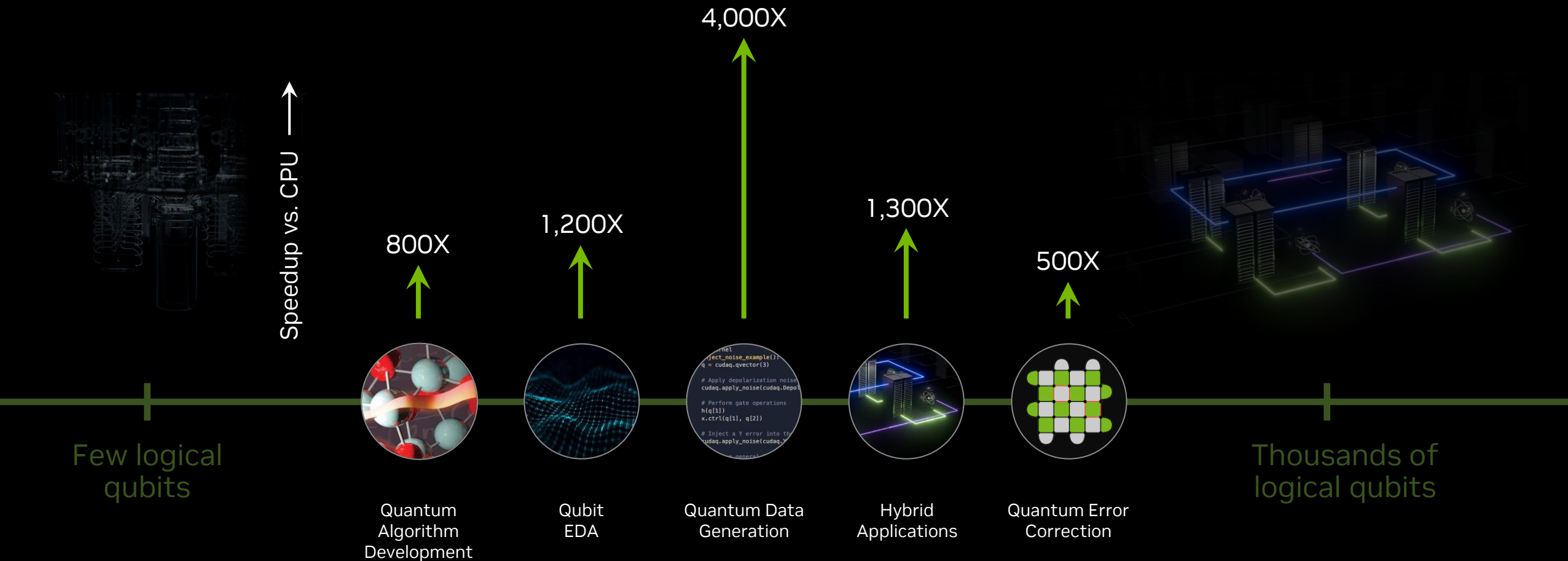


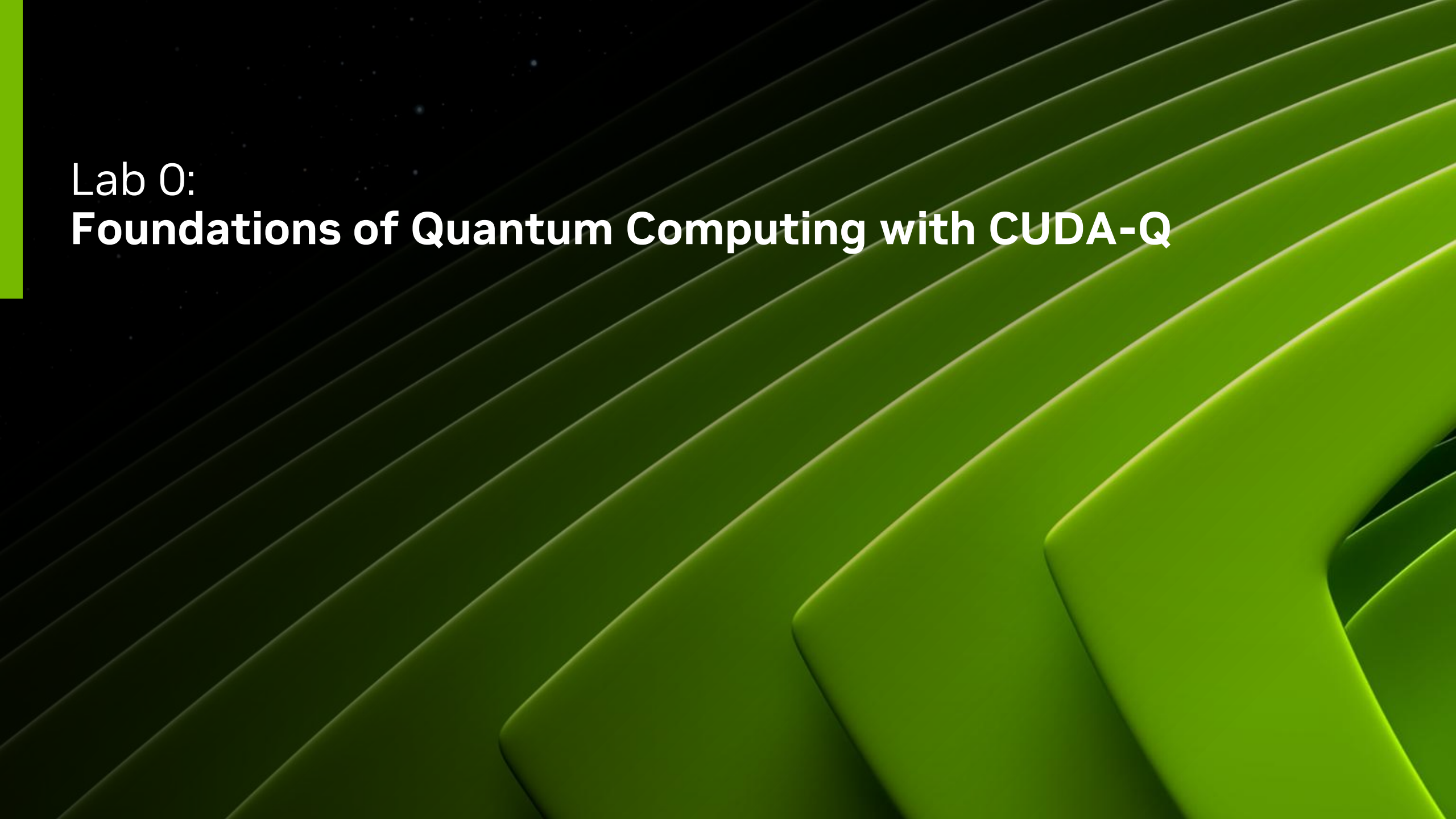
Quantum-GPU
Supercomputers

The Journey to Accelerated Quantum Supercomputing



The Journey to Accelerated Quantum Supercomputing





Lab 0: **Foundations of Quantum Computing with CUDA-Q**